

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)

Assessment Tool Weightage: 35%

Each student should simulate one question and should upload the results in Github account along with the necessary outputs and codes. This assessment tool is evaluated for 50 marks.

QUESTIONS:	Total Marks: 50	Duration: 90 Mins
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1. Simulation of 5G NR air interface performance under different numerologies, bandwidths, latency, subcarrier spacing, and throughput (BL4)
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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding Model Design	20%	Demonstrates strong understanding of underlying concepts in the simulation 5	Shows good understanding with minor gaps 4	Partial understanding; some misconceptions 3	Lacks understanding of key concepts 1-2
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution 5	Uses tools with minor errors or guidance 4	Limited ability to use tools properly 3	Unable to use simulation tools 1-2
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results 5	Analysis is reasonable with minor gaps 4	Superficial analysis; limited interpretation 3	Unable to analyze or interpret results 1-2
Documentation	20%	Presents results clearly with proper documentation 5	Documentation is adequate with minor issues 4	Poorly organized or incomplete documentation 3	No proper documentation or presentation 1-2

Criteria	Question Number :			
	Level 4	Level 3	Level 2	Level 1
Conceptual Understanding Model Design				
Tool Usage				
Analysis of Results				
Documentation				

clc;
clear;

close all;

```
%% =====  
% 5G NR AIR INTERFACE PERFORMANCE SIMULATION  
% Parameters:  
% Numerology, Bandwidth, SCS, Latency and Throughput  
% =====
```

%% 1. Simulation Parameters

% Numerology values

mu = 0:4;

% Corresponding Subcarrier Spacing

SCS = 15 * 2.^mu; % kHz

% Example bandwidths

BW = [5 10 20 40 50]; % MHz

% Modulation

modOrder = 64; % 64-QAM

bitsPerSymbol = log2(modOrder);

% Number of MIMO layers

numLayers = 2;

% Coding efficiency

codingRate = 0.85;

% Simulation duration

simulationTime = 1; % seconds

%% 2. Calculate OFDM Parameters

% Slot duration in milliseconds

slotDuration = 1 ./ (2.^mu);

% Slots per second

slotsPerSecond = 1000 ./ slotDuration;

% Approximate number of resource blocks

% Approximation based on 5G NR bandwidth

numRB = zeros(length(mu),length(BW));

```

for i = 1:length(mu)
    for j = 1:length(BW)

        % Approximate number of subcarriers
        numSubcarriers = BW(j)*1e6/(SCS(i)*1e3);

        % 12 subcarriers per RB
        numRB(i,j) = floor(numSubcarriers/12);

    end
end

%% 3. Calculate Throughput

throughput = zeros(length(mu),length(BW));

for i = 1:length(mu)

    for j = 1:length(BW)

        % Number of RBs
        RB = numRB(i,j);

        % 12 subcarriers per RB
        subcarriers = RB * 12;

        % 14 OFDM symbols per slot
        symbolsPerSlot = 14;

        % Bits transmitted per slot
        bitsPerSlot = subcarriers * symbolsPerSlot * ...
            bitsPerSymbol * codingRate * numLayers;

        % Throughput
        throughput(i,j) = bitsPerSlot * slotsPerSecond(i);

    end

end

%% Convert to Gbps
throughputGbps = throughput / 1e9;

%% 4. Latency Calculation

```

```
% Approximate transmission latency
latency = slotDuration;
```

```
%% 5. Display Results
```

```
fprintf('\n=====\\n');
fprintf('    5G NR AIR INTERFACE SIMULATION\\n');
fprintf('=====\\n\\n');
```

```
fprintf('Numerology  SCS(kHz)  Slot Duration(ms)\\n');
fprintf('-----\\n');
```

```
for i = 1:length(mu)
    fprintf('  %d      %2d      %.3f\\n', ...
        mu(i), SCS(i), slotDuration(i));
end
```

```
fprintf('\\n');
```

```
%% 6. Throughput Table
```

```
fprintf('Throughput Results (Gbps)\\n');
fprintf('-----\\n');
```

```
fprintf('Numerology  ');
```

```
for j = 1:length(BW)
    fprintf('%8d MHz',BW(j));
end
```

```
fprintf('\\n');
```

```
for i = 1:length(mu)
```

```
    fprintf('  %d      ',mu(i));
```

```
    for j = 1:length(BW)
        fprintf('%10.3f',throughputGbps(i,j));
    end
```

```
    fprintf('\\n');
```

```
end
```

```
%% =====
% 7. Plot Throughput vs Bandwidth
% =====
```

```
figure;
```

```
for i = 1:length(mu)
```

```
    plot(BW,throughputGbps(i,:),'-o','LineWidth',2);
    hold on;
```

```
end
```

```
grid on;
```

```
xlabel('Bandwidth (MHz)');
ylabel('Throughput (Gbps)');
title('5G NR Throughput vs Bandwidth');
```

```
legend('μ = 0','μ = 1','μ = 2','μ = 3','μ = 4', ...
       'Location','northwest');
```

```
%% =====
% 8. Plot Throughput vs Numerology
% =====
```

```
figure;
```

```
for j = 1:length(BW)
```

```
    plot(mu,throughputGbps(:,j),'-o','LineWidth',2);
    hold on;
```

```
end
```

```
grid on;
```

```
xlabel('Numerology (\mu)');
ylabel('Throughput (Gbps)');
title('5G NR Throughput vs Numerology');
```

```
legend('5 MHz','10 MHz','20 MHz','40 MHz','50 MHz', ...
       'Location','northwest');
```

```
%% =====
% 9. Plot Subcarrier Spacing
% =====
```

```
figure;

bar(mu,SCS);

grid on;

xlabel('Numerology (\mu)');
ylabel('Subcarrier Spacing (kHz)');
title('5G NR Numerology vs Subcarrier Spacing');
```

```
%% =====
% 10. Plot Slot Duration / Latency
% =====
```

```
figure;

bar(mu,latency);

grid on;

xlabel('Numerology (\mu)');
ylabel('Slot Duration / Approx. Latency (ms)');
title('5G NR Latency vs Numerology');
```

```
%% =====
% 11. 3D Throughput Analysis
% =====
```

```
figure;

[BWgrid,MUgrid] = meshgrid(BW,mu);

surf(BWgrid,MUgrid,throughputGbps);

xlabel('Bandwidth (MHz)');
ylabel('Numerology (\mu)');
zlabel('Throughput (Gbps)');

title('5G NR Throughput Performance');
```

```

colorbar;
grid on;

%% =====
% 12. Heatmap
% =====

figure;

imagesc(BW,mu,throughputGbps);

xlabel('Bandwidth (MHz)');
ylabel('Numerology (\mu)');
title('5G NR Throughput Heatmap');

colorbar;

%% =====
% 13. Display Latency Results
% =====

fprintf('\nLatency Results\n');
fprintf('-----\n');
fprintf('Numerology   SCS(kHz)   Latency(ms)\n');

for i = 1:length(mu)

    fprintf('   %d       %2d       %.3f\n', ...
        mu(i),SCS(i),latency(i));

end

fprintf('\nSimulation Completed Successfully.\n');

```





